E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23257967

RECEIVED 06.05.2026

ACCEPTED 22.05.2026

PUBLISHED 30-06-2026

A COMPRATIVE STUDY & ANALYSIS OF LEADING BLDC FAN BRAND IN INDIA WITH SPECIAL REFERENCE TO ATOMBERG TECHONOLOGIES PVT LTD

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ABSTRACT:

This study focuses on comparison and analysis of leading BLDC (Brush Less Direct Current) fan brands in India, with special reference to Atomberg Technologies Pvt. Ltd. In recent years, electricity consumption has increased significantly, which has created a need for energy-efficient appliances. BLDC (Brushless Direct Current) fans have emerged as an important solution because they consume less electricity while providing better performance . The main purpose of this research is to compare different BLDC fan brands based on important factors such as price, features, and customers satisfaction. The study also aims to understand consumer preferences and identify the key factors that influence their buying decisions. For this purpose, both primary and secondary data have been used. Primary data was collected through a questionnaire from 51 respondents, while secondary data was taken from company websites and other online sources. The findings of the study show that energy saving is one of the most important factors for consumers while purchasing BLDC fans. Brand reputation and price also play a major role in decision-making. Among all the brands, Atomberg is found as most preferred brand by customers due to its energy efficiency, modern features, and performance. The study concludes that BLDC fans are becoming popular in the Indian market, and consumers are slowly shifting towards energy-efficient products. It also suggests that companies should focus more on awareness, pricing strategies, and product innovation to attract more customers.

Keywords: *BLDC Fans; Atomberg Technologies; Energy Efficiency; Consumer Preferences; Customer Satisfaction; Brand Comparison; Buying Behaviour; Energy-Saving Appliances; Product Features; Indian Fan Market.*

1. Introduction

This chapter explains what other studies and research say about energy-efficient products, consumer behavior, and brand comparison. It helps to understand the basic concepts related to BLDC fans and also shows what has already been studied and what is still missing.

With the increase in electricity usage, people are becoming more aware of energy-saving products. BLDC fans are one of the important innovations in this area. However, buying decisions are not only based on technology but also on factors like price, brand image, and customer satisfaction.

1.1 Energy-Efficient Appliances

Energy-efficient appliances are those products which use less electricity but give good performance. Many studies have shown that using such products helps in reducing electricity bills and saving energy.

BLDC fans are a good example of energy-efficient appliances. They consume almost 50 to 60% less electricity compared to traditional fans. Because of this, they are becoming popular among consumers and people .

However, some studies also show that people hesitate to buy these products because of their higher price and lack of awareness. Many consumers focus on short-term cost instead of long-term savings.

1.2 Consumer Buying Behaviour

Consumer buying behavior means how people decide what to buy and why they buy it. It depends on many factors such as price, quality, brand name, and personal preference.

In India, many consumers are price-sensitive, which means they prefer products that are affordable. But nowadays, people are also starting to consider energy saving and product features while buying.

Research shows that when people understand the long-term benefits, they are more likely to buy energy-efficient products like BLDC fans.

1.3 Brand Competition and Positioning

In the BLDC fan market, many companies are competing with each other. Each brand tries to create a different image in the minds of customers.

Some brands focus on technology and innovation, while others focus on affordability. For example, Atomberg is known for energy efficiency and modern features, while other brands may focus on strong brand reputation or pricing.

Studies show that a strong brand image increases customer trust and makes it easier for people to choose a product.

1.4 Pricing Strategy

Price is one of the most important factors in buying decisions. Many consumers compare the price of different brands before making a purchase.

BLDC fans are usually more expensive than traditional fans, which can be a barrier for some customers. However, if customers understand that they will save money on electricity in the long run, they may be willing to pay a higher price.

Research shows that customers look for value for money, which means they compare price with the benefits they receive.

1.5 Product Features and Technology

Modern consumers prefer products that offer additional features. BLDC fans come with many advanced features such as:

1.5.1 Remote control

1.5.2 Low noise operation

1.5.3 Energy saving mode

1.5.4 Smart connectivity (in some models)

These features make the product more attractive to customers.

However, some customers still prefer simple products without too many features, especially if they are cheaper.

1.6 Customer Satisfaction

Customer satisfaction means how happy customers are with a product. It depends on performance, quality, price, and service.

If a product meets or exceeds customer expectations, then satisfaction is high. Satisfied customers are more likely to recommend the product to others.

In the case of BLDC fans, satisfaction depends on factors like energy saving, air delivery, noise level, and durability.

1.7 Relationship Between Price, Features, and Satisfaction

Many studies show that there is a strong connection between product features, price, and customer satisfaction.

If a product offers good features at a reasonable price, customers feel satisfied. On the other hand, if the price is high and the features are not good, satisfaction becomes low.

So, companies need to balance both price and features to attract customers.

1.8 Research Gap

Even though many studies have been done on energy-efficient products and consumer behavior, there are still some gaps:

1.8.1 Most studies focus only on technical aspects

1.8.2 Very few studies compare different BLDC fan brands

1.8.3 Limited research is available on customer satisfaction in this segment

1.8.4 Lack of studies combining price, features, and satisfaction together

This study tries to fill these gaps by comparing different BLDC fan brands and understanding customer preferences.

1.9 Conclusion of Literature Review

From the above discussion, it is clear that buying decisions depend on many factors like price, features, brand image, and satisfaction.

BLDC fans are gaining popularity because of their energy-saving benefits, but their adoption is still growing. This shows that more awareness and better comparison of brands is needed.

This study helps in understanding these factors and provides a clear comparison of leading BLDC fan brands in India.

2. RESEARCH METHODOLOGY

2.1 Introduction

This chapter explains the method used to conduct the research. It shows how the data was collected, analyzed, and interpreted. A proper research method helps in getting accurate and reliable results.

In this study, the focus is on comparing different BLDC fan brands and understanding consumer preferences.

2.2 Research Design

The study uses a **descriptive research design**.

Descriptive research is used to describe and understand the current situation. It helps in analyzing consumer preferences, brand comparison, and satisfaction levels.

This design is suitable because the study focuses on:

- Comparing different brands
- Understanding customer behavior
- Analyzing satisfaction

2.3 Research Approach

The study follows a **quantitative research approach**.

This approach involves collecting numerical data and analyzing it using simple statistical methods. It helps in identifying patterns and trends in consumer responses.

2.4 Data Sources

The study uses both **primary data** and **secondary data**.

Primary data was collected directly from respondents using a structured questionnaire.

2.4.1 Data collected from **51 respondents**

2.4.2 Google Form was used for data collection

2.4.3 Respondents include students, professionals, and business owners

This data helps in understanding real customer opinions. Secondary data was collected from:

2.4.4 Company websites

2.4.5 Online platforms (like Amazon, Flipkart reviews)

2.4.6 Research articles and reports

This data provides background information and supports the study.

2.5 Sampling Technique

The study uses **convenience sampling**.

This means respondents were selected based on availability and ease of access. It is commonly used in academic research because it is simple and time-saving.

2.6 Sample Size

The total sample size is **51 respondents**. These respondents include:

- 2.6.1 Students
- 2.6.2 Working professionals
- 2.6.3 Business owners
- 2.6.4 General consumers

This sample size is enough to understand basic patterns and trends.

2.7 Data Collection Instrument

A **structured questionnaire** was used to collect data. The questionnaire includes:

1. **Demographic Information**
   - Age
   - Occupation
2. **Product Usage**
   - Whether they use BLDC fans
   - Brand used
3. **Buying Behaviour**
   - Price range
   - Purchase factors
4. **Satisfaction Level (Rating 1–5)**

- Energy efficiency
- Air delivery
- Noise level
- Design
- Value for money

5. **Consumer Preference**
    - Brand importance
    - Recommendation

2.8 Variables of the Study

The study includes the following variables:

Independent Variables:

- Product features and pricing

Mediating Variable:

- Consumer perception

Dependent Variable:

- Customer satisfaction

These variables help in understanding how different factors affect customer satisfaction.

2.9 Data Analysis Techniques

The data was analyzed using simple statistical methods:

2.9.1 Percentage analysis

2.9.2 Average calculation

2.9.3 Comparison between brands

Tools used:

2.9.4 Microsoft Excel

2.9.5 Google Sheets

Data is presented using:

2.9.6 Tables

2.9.7 Pie charts

2.9.8 Bar graphs

2.10 Conceptual Model

The study follows this simple model:

Product Features & Pricing → Consumer Perception → Customer Satisfaction

This means:

2.10.1 Features and price affect how customers think

2.10.2 This thinking affects their satisfaction

2.11 Limitations of the Study

There are some limitations:

2.11.1 Sample size is small (51 respondents)

2.11.2 Only limited areas are covered

2.11.3 Responses are based on personal opinion

2.11.4 Not all brands are included

But still, the study gives useful insights.

3. DATA ANALYSIS & INTERPRETATION

3.1 Introduction

This chapter presents the analysis of data collected from 51 respondents through a questionnaire. The main aim is to understand consumer preferences, buying behaviour, and satisfaction levels related to different BLDC fan brands.

The data is analyzed using percentage method and simple comparison so that it is easy to understand.

3.2 Demographic Analysis

1. **Age Group**

Most of the respondents belong to the **36–50 age group**, followed by the **18–25 age group**. Very few respondents are above 50 years.

This shows that middle-aged people are more involved in purchasing decisions, while young people are also becoming aware of BLDC fans.

2. **Occupation**

The majority of respondents are **students and business owners**, followed by working professionals.

This indicates that both young consumers and business-minded individuals are interested in energy-saving products.

3.3 Product Usage Analysis

3.3.1 Ownership of BLDC Fans

Almost all respondents (around **96%**) already use BLDC fans.

This shows that awareness and usage of BLDC fans is quite high among the respondents.

3.3.2 Brand Used

Most respondents (around **60%**) use **Atomberg Technologies Pvt. Ltd.** fans. Other brands like:

- Crompton Greaves Consumer Electricals
- Havells India Ltd
- Orient Electric

have comparatively lower usage.

This clearly shows that Atomberg is the most popular brand among the respondents.

3.3.3 Number of Fans Owned

Most respondents own **2–3 BLDC fans**, and many even own **more than 4 fans**.

This indicates that people are not just trying BLDC fans, but are using them widely in their homes.

3.3.4 Duration of Usage

Most users have been using BLDC fans for **1–2 years**, while many have used them for more than 2 years.

This shows that BLDC fans are not new for consumers and they have good experience with them.

3.4 Purchase Behaviour Analysis

1. Place of Purchase

Most respondents purchased their fans from **offline stores**.

Some purchased from dealers or distributors, while very few bought online. This shows that customers still prefer physical stores for such products.

2. Price Range

Most people purchased fans in the range of **₹2,500–₹3,500**, followed by ₹3,500–₹4,500. A smaller group purchased premium fans above ₹4,500.

This shows that customers prefer **affordable and mid-range products**.

3. Factors Influencing Purchase

The most important factor is **brand reputation**, followed by:

- Price
- Energy saving
- Features
- Design

This means customers trust well-known brands more than just features or design.

4. Brand Comparison

All respondents said they compared multiple brands before purchasing.

This shows that customers are very careful and do proper research before buying.

5. Importance of Brand Reputation

Most respondents said brand reputation is **very important**.

This confirms that trust and brand image play a big role in decision-making.

a. Satisfaction Analysis

6. Energy Efficiency

Most respondents rated energy efficiency as **4 or 5 out of 5**.

This shows that energy saving is the biggest strength of BLDC fans.

7. Air Delivery (Performance)

Most respondents gave a rating of **4 or 5**, which means they are satisfied with airflow and performance.

8. Noise Level

Most users are satisfied with the **low noise level** of BLDC fans. This shows that silent operation is an important advantage.

9. Overall Observation

From the data, it is clear that:

a. Customers are highly satisfied with BLDC fans
b. Energy saving is the main reason for purchase
c. Atomberg is one of the most preferred brand
d. Price and brand trust also influence decisions

e. Key Findings

From the analysis, the following key points are identified:

f. Atomberg is the most popular BLDC fan brand
g. Most customers prefer mid-range pricing
h. Brand reputation is the most important factor
i. Energy efficiency is highly valued
j. Customers compare multiple brands before buying
k. Overall satisfaction level is high

4. Conclusion

This study focused on comparing leading BLDC fan brands in India with special reference to Atomberg Technologies Pvt. Ltd.. The main aim was to understand consumer preferences, compare different brands, and analyze factors affecting buying decisions.

From the analysis, it is clear that BLDC fans are becoming popular because they consume less electricity and provide better performance. People are slowly shifting from traditional fans to energy-efficient options.

The study shows that **energy saving is the most important factor** influencing customer decisions. Along with this, **brand reputation and price** also play a major role. Customers prefer brands they trust and products that offer good value for money.

Among all the brands, Atomberg has emerged as the most preferred brand. It performs well in terms of energy efficiency, features, and customer satisfaction. Other brands like Havells India Ltd, Crompton Greaves Consumer Electricals, and Orient Electric are also present in the market but have lower preference compared to Atomberg in this study.

The study also shows that customers are becoming more aware and they compare multiple brands before making a purchase. Overall, the satisfaction level among users is high, which indicates a positive future for BLDC fans in India.

5.1 Suggestions

Based on the findings of the study, the following suggestions can be given:

For Companies

- Companies should focus more on **creating awareness** about energy-saving benefits
- Pricing should be made more affordable to attract more customers
- Brands should improve **after-sales service** to increase trust
- More innovative features can be added to attract young customers

For Atomberg

- Atomberg should maintain its strong position in energy efficiency
- It can expand its market by targeting rural and semi-urban areas
- Competitive pricing strategies can help increase market share
- More marketing and promotion can increase brand awareness

For Other Brands

- Other brands should focus on improving product features and innovation

- They should work on building stronger brand trust
- Offering competitive pricing can help attract more customers

For Consumers

- Consumers should focus on long-term savings instead of only initial price
- They should compare features and performance before buying
- Awareness about energy-efficient products should be increased

5.2 Limitations of the Study

- The sample size is limited to 51 respondents
- The study is limited to selected brands only
- Responses are based on personal opinions
- The study is limited to a specific area

5.3 Scope for Future Study

- Future research can be done with a larger sample size
- More brands can be included for better comparison
- Detailed technical analysis can also be added
- Studies can be conducted in different cities

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